

Bugs found: 5

1. BadKey overrides equals(Object) but does not override hashCode(), violating the equals/hashCode contract.
2. BadKey.equals(Object) returns true for null, violating the equals contract.
3. BadKey.equals(Object) returns true for objects of unrelated types, violating the equals contract.
4. Calling k.hashCode() when k is null causes a NullPointerException.
5. Assigning a[-1] = 10 uses an invalid negative array index and would cause an ArrayIndexOutOfBoundsException.